

Ayush Nikhade

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Summary

Full-Stack & AI/ML Engineer with production experience building scalable microservices, real-time IoT/telemetry architectures, and deep learning pipelines. Strong foundation in Data Structures, Algorithms, and distributed application security.

Experience

UPTIQ.ai Remote
Software Development Engineer Intern (Loan Origination System) 6 Months

- Engineered production workflows for a fintech Loan Origination System (LOS), delivering multi-party intake (co-borrower & guarantor invites, KYC verification, ownership checks, and submission gating).
- Integrated third-party APIs including J.D. Power vehicle valuations, Census geocoding (MSA, tract, county), and Cobalt OFAC/UCC/SOS risk intelligence screening into borrower and broker pipelines.
- Implemented server-side underwriting pipelines to calculate Debt-to-Income (DTI), personal liabilities, and net worth from disclosures.
- Hardened application security by resolving stored-XSS vulnerabilities, securing sensitive endpoints, and enforcing typed API contract layers.

Projects

Crowd Sync – AI-Powered Bus Fleet Management System [Code]

Technologies: React, Node.js, Express, Socket.IO, PostgreSQL, Redis, Python, YOLOv8, OpenCV, Leaflet.js

- Built an end-to-end IoT and fleet platform connecting edge AI devices to a real-time web dashboard for live bus tracking and passenger density analytics.
- Engineered a Python edge pipeline using YOLOv8 and OpenCV to calculate occupancy levels and stream telemetry/GPS over WebSockets.
- Architected a high-throughput Node.js backend backed by PostgreSQL and Redis caching to ingest telemetric data streams with sub-second latency.
- Developed a monitoring dashboard in React with Leaflet.js for live route mapping, congestion indicators, and fleet performance metrics.

Satellite Vision – Geospatial Land Cover & Change Detection [Code]

Technologies: Python, FastAPI, TensorFlow, OpenCV, NumPy, MongoDB, React, Vite

- Built an end-to-end computer vision platform utilizing an 8-class EuroSAT CNN to classify gigapixel satellite imagery and detect temporal land cover changes (urbanization, deforestation, water depletion).
- Implemented an overlapping sliding-window tiler (64×64 with 50% stride) to eliminate GPU OOM errors, with probability-based confidence stitching.
- Applied OpenCV morphological filtering to eliminate false-positive noise and isolate genuine ecological transitions between temporal image pairs.

Technical Skills

Languages: C++ , Python, JavaScript, TypeScript, Java, SQL, HTML/CSS

Frontend: React.js, Next.js, Vite, Tailwind CSS, Leaflet.js, Framer Motion

Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets (Socket.IO), JWT

Databases & Caching: PostgreSQL, MongoDB, MySQL, Redis

AI, ML & CV: YOLOv8, OpenCV, TensorFlow, Keras, Scikit-learn, Computer Vision

Tools & Cloud: Git, GitHub, Docker, Postman, Vercel, VS Code

Education

Walchand College of Engineering, Sangli Sangli, India
B.Tech in Computer Science 2022 – 2026
CGPA: 8.4 / 10.0

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Object-Oriented Programming, Machine Learning, Deep Learning

Achievements & Certifications

Competitive Programming: Solved 300+ algorithmic problems across LeetCode, CodeChef, and GeeksforGeeks.

CDAC Certification: Certified in Ethical Hacking and Penetration Testing.